

George Townsend

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Geodesy researcher developing inverse methods and open-source software for remote sensing of water level and soil moisture. Working to further establish GNSS-IR as an open tool for environmental monitoring.

Interests: GNSS reflectometry, Inverse problems, Signal processing, Time-series analysis, Error characterization

Education

University of Newcastle – Callaghan, NSW, Australia

- Master of Philosophy (Electrical Engineering)
- Bachelor of Surveying – First Class Honours

Jul 2025 – Apr 2027 (expected)

Jan 2021 – Jun 2025

Research & Professional Experience

Research Assistant, University of Newcastle – Callaghan, NSW, Australia

Aug 2026 – Present, Dec 2024 – Jan 2026

- Identified and characterized a sidereal aliasing artifact affecting sub-daily GNSS-IR soil moisture estimation
- Designed and constructed a solar-powered GNSS-IR system for remote soil moisture monitoring
- Led the deployment of the GNSS-IR monitoring system to Warkworth Space Centre, New Zealand

Visiting Scholar, The Ohio State University – Columbus, OH, USA

Feb 2026 – Aug 2026

- Developed a penalized-spline inverse-SNR method to increase temporal resolution of GNSS-IR
- Produced novel observations of tsunami induced sea surface oscillations at periods less than 30 minutes

Spatial Data Controller, Anditi – Newcastle, NSW, Australia

Dec 2021 – Dec 2024

- Performed manual and automated QA for multi-terabyte aerial LiDAR classification, including Python scripting

Open Source Contributions

gnssrefl

github.com/kristinemlarson/gnssrefl

- Widely-used Python package maintained by Professor Kristine Larson for GNSS-IR research
- Responsible for major features including soil moisture estimation from GLONASS, Galileo, and Beidou signals
- Refactored and vectorized core algorithms, replacing legacy Fortran with NumPy (up to 20x faster)

Scholarships

Research Training Program (RTP) Scholarship

- Full tuition fee offset and stipend to support postgraduate research in GNSS signal processing

Selected Publications & Presentations

- Townsend, G., Han, S.-C., Larson, K., & Yeo, I.-Y. “Towards Sub-daily GNSS-IR Soil Moisture Estimation: The Sidereal Drift Artifact.” *GPS Solutions*. Manuscript under review.
- Townsend, G., Han, S.-C., Roger, J., & Okal, E. “Zero-Configuration GNSS Interferometric Reflectometry Resolves 20-Minute Tsunami-Coda Oscillations.” *AGU*, San Francisco, CA, USA. Abstract accepted.
- Townsend, G., Han, S.-C., Larson, K., & Yeo, I.-Y. “Towards Sub-daily GNSS-IR Soil Moisture Estimation.” *EGU General Assembly 2026*, Vienna, Austria. Poster abstract accepted; not presented.
- Wilson, M., Han, S.-C., Khaki, M., Yeo, I.-Y., Townsend, G., Yuan, F., Nguyen, M., Cai, X., Ash, C., & Moller, D. “Satellite Sensing into Agricultural Practices: Cal/Val Campaign of Satellite, Airborne and Ground GNSS-Reflectometry of Soil Moisture.” *IGARSS 2025*, Brisbane, Australia.
- Townsend, G. “Measuring Soil Moisture With GPS Multipath and Open Source Software.” *FOSS4G 2025*, Auckland, New Zealand.